

SMSE: Social-Media Stock Estimator

`github/social-media-stock-estimator` — `figma/social-media-stock-estimator`

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Abstract—Online investment communities increasingly influence short-term market dynamics, motivating the need to understand whether social-media sentiment contains actionable predictive structure. This work presents the Social-Media Stock Estimator (SMSE), an end-to-end system that integrates Reddit-derived sentiment with financial data to forecast 7-day stock price movements. The pipeline combines large-scale Reddit collection, text preprocessing, FinBERT-based sentiment extraction, market-data alignment, and multimodal feature construction for regression modeling. We develop and evaluate several model variants to assess the predictive value of sentiment, semantic embeddings, and metadata features, and we conduct exploratory analysis to characterize the strength and stability of sentiment–price relationships. In addition, we design an interactive prototype interface that demonstrates how such predictions can be surfaced to users in a transparent, interpretable manner. Overall, the study assesses the feasibility of sentiment-informed short-horizon forecasting and highlights the methodological and design considerations required for deploying such systems responsibly.

I. INTRODUCTION

The rise of social media as a dominant force in financial markets has fundamentally altered the dynamics of price discovery and investor behavior, exemplifying how information networks shape collective decision-making in high-stakes domains. Online communities such as *r/wallstreetbets*, *r/stocks*, and *r/investing* have demonstrated unprecedented collective influence, most notably during the 2021 GameStop episode, where coordinated retail investor activity driven by social network effects drove extreme market volatility. These platforms transform individual sentiment into collective action at scales and speeds previously unattainable, creating observable effects on short-term stock price movements through network cascades and information contagion. This project introduces the Social-Media Stock Estimator, a machine learning system designed to forecast 7-day percentage changes in stock prices by integrating social media sentiment analysis with traditional market data. The work is motivated by fundamental questions in social and information networks: how does information propagate through online communities, how do network structures influence collective behavior, and can aggregated signals from distributed actors predict real-world outcomes? By analyzing over 5,200 Reddit posts and 80,000 comments across 18 companies and 4 subreddits, the system validates that aggregated sentiment—when properly

filtered, temporally aligned, and combined with market indicators—contains measurable predictive signals about short-term price movements.

Understanding social media’s influence on markets matters because it represents a fundamental shift in information flow and market efficiency mediated by network topology and community dynamics. Unlike traditional news or analyst reports that follow hierarchical broadcast models, social media discussions occur through peer-to-peer information exchange in networked communities, evolve rapidly through cascading interactions, and reflect the collective psychology of millions of retail investors whose connections and information-sharing behaviors create emergent market signals. While institutional investors employ sophisticated sentiment analysis tools, retail investors lack accessible, interpretable systems that translate social sentiment and network activity into decision-support insights. The democratization of market participation through zero-commission trading platforms has amplified this information asymmetry while simultaneously strengthening the network effects that enable coordinated investor behavior, making social sentiment a critical variable in modern market dynamics with direct implications for risk management, portfolio optimization, and regulatory oversight. This project explores whether community-level signals extracted from social networks—considering factors like discussion volume, sentiment coherence across subreddits, and temporal patterns of information propagation—provide predictive value beyond individual post sentiment. Moreover, survey evidence reveals that potential users express skepticism about “black box” sentiment predictions while valuing transparency and source provenance, highlighting that technical accuracy alone is insufficient for practical adoption in systems that leverage social network data.

However, predicting stock movements from social media presents fundamental challenges that reflect broader issues in social and information networks research. First, social media platforms are inherently noisy environments where valuable insights coexist with spam, misinformation, and coordinated manipulation, requiring sophisticated filtering beyond simple sentiment scoring—a problem analogous to identifying signal versus noise in large-scale network data. Second,

discussions occur asynchronously and influence markets over varying time horizons through complex diffusion processes, making temporal alignment and decay modeling critical but difficult to tune without overfitting. Third, predictive value varies dramatically across network structures and community norms—during meme stock episodes characterized by dense interconnection and rapid information spread, social media becomes a primary price driver, while in normal conditions with weaker network effects fundamental factors dominate. Fourth, representing textual discussions as numerical features requires balancing high-dimensional transformer embeddings against interpretability and overfitting risks, particularly when network-level features (subreddit membership, cross-community sentiment patterns) must be integrated with post-level signals. Fifth, not all users are equally informative, and active Reddit users are not representative of the broader investment community, introducing sampling bias characteristic of many social network studies where observed participants differ systematically from the underlying population.

II. RELATED WORK

The integration of social media sentiment into stock price prediction has evolved from simple aggregate sentiment scoring to sophisticated multi-modal approaches that capture conversation dynamics, temporal decay, and regime-specific behavior. Early approaches relied on lexicon-based sentiment tools like VADER and TextBlob [1], [2], but recent work has demonstrated the superiority of transformer-based models for financial text classification. Ferraro and Sperli [3] show that RoBERTa-based sentiment extraction from Reddit and Twitter substantially improves prediction accuracy during the 2021 GameStop episode, while Awan et al. [4] note that classical sentiment classifiers achieve only moderate accuracy compared to domain-specific transformer approaches. These findings motivated our adoption of FinBERT, a BERT variant fine-tuned specifically for financial sentiment, which better captures the nuanced language of investment discussions than general-purpose models.

Beyond sentiment extraction methods, a critical insight from recent literature is that aggregate sentiment alone provides insufficient signal—conversation-level features and user quality substantially improve predictive power. Li et al. [5] demonstrate that conversation-level measures including argument similarity, sentiment similarity, and discussion size predict abrupt weekly returns ($\pm 20\%$) in IPO firms, suggesting that coordination and echo-chamber dynamics drive price movements beyond simple bullish-bearish ratios. Similarly, Coyne et al. [6] show that filtering for “smart users” with historically accurate forecasts raises StockTwits prediction accuracy by approximately 15%, though this approach requires extensive user histories and suffers from small sample sizes after filtering. These insights inform our feature engineering strategy, which incorporates sentiment strength, confidence ratios, and bull-bear interaction terms

to capture conversation quality without requiring individual user tracking—a practical consideration given Reddit’s pseudonymous structure and our focus on post-level rather than user-level analysis.

Temporal modeling presents another critical design choice that existing work addresses inconsistently. Wang et al. [7] introduce explicit sentiment-effect parameters and decay modeling over rolling windows, finding optimal performance at six days for DJIA prediction using news sentiment, which aligns closely with our 7-day forecasting horizon. However, their uniform decay assumption may not capture Reddit’s more volatile sentiment cycles. Costola et al. [8] address this through regime-switching models that identify synchronized cointegration among sentiment, prices, and volumes during meme stock episodes using Bayesian MCMC methods. While their econometric rigor provides valuable insights into extreme coordination events, the approach operates retrospectively and focuses on identifying meme periods post hoc rather than providing actionable next-day or next-week forecasts applicable to normal market conditions. Our temporal feature engineering balances these considerations by incorporating both cyclical time encodings and sentiment-price interaction terms that allow the model to adapt to varying market regimes without explicit regime classification.

The choice of modeling architecture involves trade-offs between expressiveness and interpretability that the literature highlights through mixed empirical results. Deep sequence models like LSTMs and GRUs have shown promise in capturing temporal dependencies [1]–[3], with Ouf et al. reporting strong R^2 improvements when combining LSTM-processed Twitter sentiment with historical prices for Apple, Google, and Tesla. However, Awan et al. [4] find that tree-based models underperform on high-dimensional text embeddings without systematic feature engineering, suggesting that model complexity alone does not guarantee performance. Our experiments comparing ElasticNet regression against LightGBM gradient boosting on PCA-reduced FinBERT embeddings support this finding—regularized linear models with carefully engineered features (sentiment strength, confidence, bull-bear ratios, sentiment-price interactions) substantially outperform ensemble methods on our dataset.

Dataset composition and coverage also significantly influence generalizability. Several studies focus on narrow contexts: Ferraro and Sperli [3] analyze only GameStop during its 2021 meme episode, Costola et al. [8] examine three meme stocks (GME, AMC, KOSS), and Li et al. [5] restrict analysis to IPO firms, limiting applicability to normal market regimes and established companies. Multi-source approaches that combine platforms like Twitter, Yahoo! Finance message boards, and news [2] offer broader coverage but introduce heterogeneity in discussion norms and sentiment expression that complicates aggregation. Our dataset addresses

these limitations by collecting a large number of posts and comments across 18 companies spanning multiple sectors (technology, finance, energy, automotive, consumer goods) from four major Reddit investment communities (r/stocks, r/investing, r/wallstreetbets, r/StockMarket), providing sufficient diversity to evaluate performance across normal market conditions while maintaining platform consistency.

Synthesizing these insights, effective social sentiment prediction systems require four key components that existing work addresses individually but rarely integrates systematically: (1) transformer-based sentiment extraction that captures financial domain language, (2) conversation-level features beyond aggregate sentiment that reflect coordination dynamics, (3) explicit temporal modeling through appropriate prediction windows and decay functions, and (4) careful dimensionality reduction and feature engineering to prevent overfitting on high-dimensional text representations.

III. METHODS

This project consists of two complementary components: (1) a full end-to-end **data ingestion and modeling pipeline** that transforms raw Reddit discussions into sentiment-aligned financial signals, and (2) a high-fidelity **Figma prototype** illustrating how end users would interact with sentiment-driven forecasts in a professional stock-analysis interface.

Together, these components demonstrate both the technical feasibility and the user-centered design of the Social-Media Stock Estimator (SMSE) system.

A. Data Ingestion and Processing Pipeline

The pipeline is designed to be modular, reproducible, and deployable. Fig. 1 shows the overall architecture, which spans Reddit crawling, preprocessing, sentiment labeling, price augmentation, and multimodal feature construction for downstream regression modeling.

Beyond constructing the technical pipeline, this stage of the project serves an important exploratory purpose: to evaluate whether real-world Reddit discussion exhibits measurable and modelable correlation with short-term stock price movements. Because retail-driven sentiment is noisy, heterogeneous, and highly context-dependent, an empirical assessment is necessary before investing in full system development. The preprocessing, EDA, and baseline modeling steps therefore function as a feasibility study—testing whether sentiment extracted from authentic Reddit posts contains sufficient signal, when aligned with financial data, to support predictive modeling. These analyses allow us to quantify the strength, stability, and limitations of sentiment–price relationships in practice, and to determine whether building a larger-scale, real-time sentiment forecasting system is justified.

1) *Data Collection (Reddit Crawler)*: We implemented a PRAW-based crawler that systematically fetches discussion posts from four investment-related subreddits (r/stocks, r/StockMarket, r/investing, r/wallstreetbets) for 18 large-cap companies.

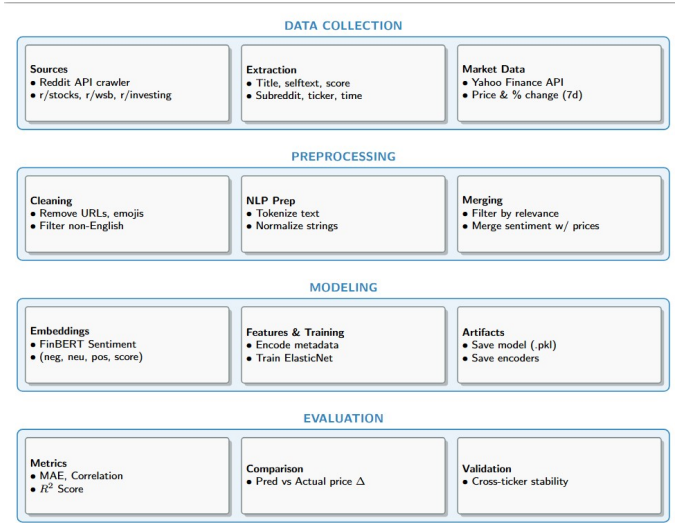


Fig. 1. Architecture of SMSE model ingestion pipeline

For each (company, subreddit) pair, the crawler collects approximately 120 top posts from the past month, extracting:

- Post ID, title, selftext, score, comment count
- Timestamps and metadata
- All top-level comments
- Ground-truth company/ticker

This produced a v2 dataset of **5,206 posts** and **80,141 comments**, summarized in Table I. This large-scale dataset provides the empirical foundation needed to evaluate whether sentiment signals embedded in real Reddit discussions can be reliably aligned with financial outcomes and modeled for prediction.

TABLE I
REDDIT DATA SUMMARY (V2)

Metric	Value
Total Posts	5,206
Total Comments	80,141
Companies	18
Subreddits	4
Avg. Comments/Post	15.4
Data Version	v2

2) *Preprocessing and Feature Engineering*: Once raw Reddit activity is collected, the next stage standardizes the text, extracts sentiment, and aligns posts with market data to construct a clean modeling-ready dataset.

The preprocessing and feature engineering pipeline converts raw Reddit posts into a structured multimodal dataset for regression modeling. It consists of five stages: data consolidation, text normalization, sentiment inference, price augmentation, and feature construction.

Raw Data Consolidation. All JSON files from the Reddit crawler are merged into a unified dataset (5,200 posts) containing title, selftext, score, subreddit, ticker, and timestamp. Duplicate and malformed entries are removed.

Text Cleaning and Normalization. Posts are cleaned through URL removal, symbol and whitespace normalization, lowercasing, and title–body concatenation. Empty or non-English posts are discarded. NLTK tokenization and stopword removal provide auxiliary linguistic structure.

Sentiment Inference with FinBERT. Each cleaned post is fed through the FinBERT model to obtain negative, neutral, and positive sentiment probabilities, a discrete label (1/0/1), and a continuous sentiment score:

$$\text{sentiment_score} = p_{\text{pos}} - p_{\text{neg}}.$$

FinBERT’s 768-dimensional CLS embedding is also extracted as the semantic representation of the post.

Price Augmentation (YFinance). Yahoo Finance is used to retrieve the stock’s closing price on the posting date and seven days later. The target variable is the forward percentage return:

$$\text{price_pct_change} = \frac{p_{t+7} - p_t}{p_t} \times 100.$$

Feature Engineering. To prepare the data for multimodal regression, several feature groups are constructed:

- **PCA Reduction:** FinBERT embeddings (768D) are reduced to 128D using PCA to limit noise and dimensionality.
- **Sentiment Features:** Sentiment probabilities, sentiment score, and sentiment confidence ($|p_{\text{pos}} - p_{\text{neg}}|$).
- **Metadata Encoding:** One-hot encodings for ticker and subreddit; normalized post score.
- **Temporal Features:** Cyclical encodings (e.g., $\sin/\cos$ day-of-week) for posting time.
- **Interaction Features:** Cross-terms such as sentiment–price interactions and bull–bear ratios capturing behavioral effects.

The resulting dataset combines linguistic, sentiment, metadata, and temporal signals in a compact numerical representation suitable for regression modeling.

3) *Regression Modeling Pipeline:* With the transformed dataset, the modeling pipeline integrates semantic, sentiment, and metadata features to evaluate whether Reddit-derived signals carry predictive value for short-term price movement.

The modeling pipeline fuses textual, sentiment, and metadata components:

- FinBERT CLS embeddings (768D)
- PCA dimensionality reduction ($768 \rightarrow 128$)
- Sentiment features (pos/neu/neg probabilities, sentiment score)
- Metadata (subreddit, ticker, post score, temporal encodings)
- Engineered interactions (bull–bear ratio, sentiment $\times$ price terms)

We train an **ElasticNetCV** model, balancing sparsity and stability through a mixture of L_1 and L_2 penalties. PCA mitigates noise and overfitting while significantly improving model interpretability.

B. Figma Prototype and User Interaction Design

To complement the pipeline, we designed a high-fidelity Figma prototype (Fig. 3) demonstrating how end users interact with forecasts, sentiment visualizations, and interpretability tools.

1) *Authentication and User Onboarding:* The prototype begins with a modern authentication flow including email/password signup, login, and Google OAuth. This mirrors real-world financial apps requiring personalized watchlists and user-specific insights.

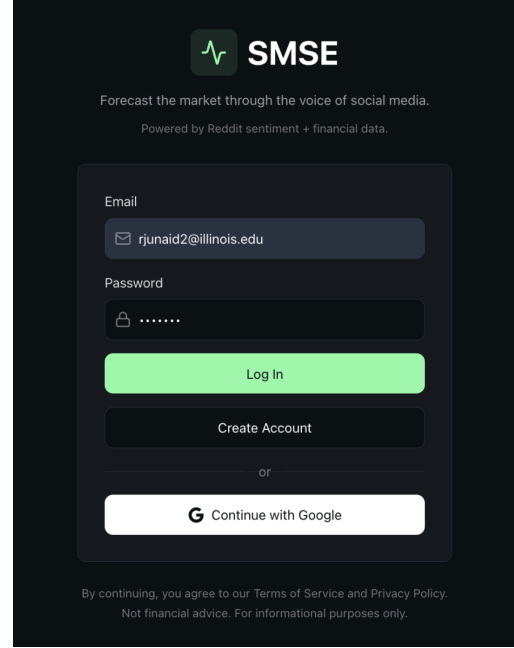


Fig. 2. SMSE Figma Prototype: Login screen.

2) *Dashboard and Watchlist:* Users land on a dashboard summarizing:

- Predicted 7-day returns for watchlist stocks
- Sentiment distributions
- Trend indicators for high volatility or high-comment-volume tickers
- Reddit “Trending” modules showing active discussions

This aligns model outputs with intuitive financial UI patterns.

3) *Ticker Detail View:* The ticker-level interface is organized into three coordinated views that surface model outputs, sentiment dynamics, and explanatory signals in an interpretable manner.

Overview Tab. The Overview tab summarizes the aggregate sentiment landscape for the selected ticker. A sentiment-distribution chart shows the proportion of positive, neutral, and negative posts, while a “Key Insights” panel highlights notable discussion themes such as earnings expectations, investor interest, or valuation concerns. A sentiment-over-time chart visualizes daily mood trends leading up to the prediction window.

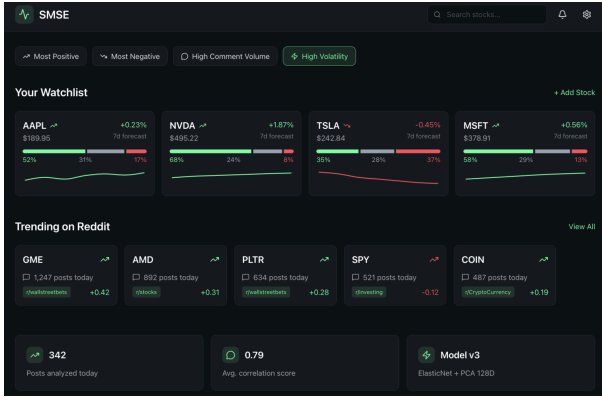


Fig. 3. SMSE Figma Prototype: Dashboard with watchlist, trending stocks, and statistics.

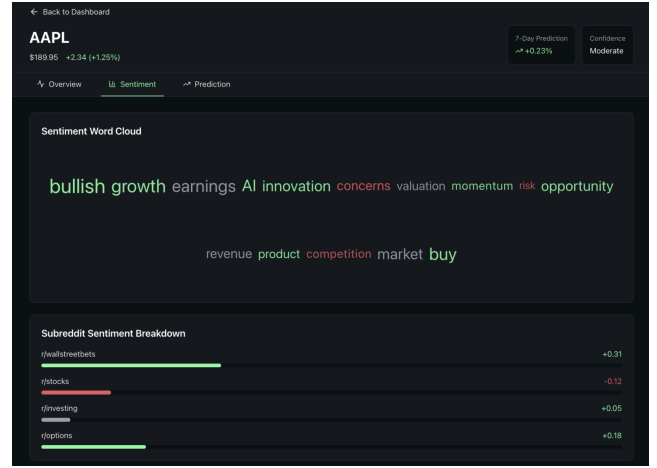


Fig. 5. Sentiment tab: word cloud and subreddit-level sentiment comparison.

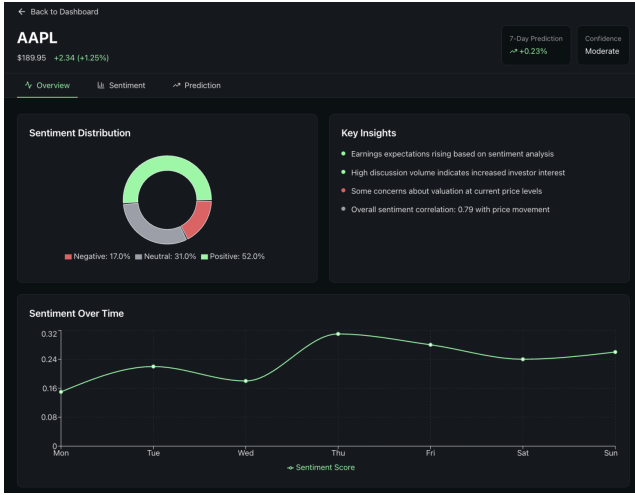


Fig. 4. Overview tab: sentiment distribution, key insights, and sentiment-over-time visualization.

Sentiment Tab. The Sentiment tab provides a qualitative and community-level breakdown of discussion patterns. A FinBERT-generated word cloud highlights salient terms associated with positive or negative sentiment, and subreddit-level sentiment bars compare tone across communities (e.g., r/wallstreetbets, r/stocks, r/investing), enabling users to assess where optimism or skepticism is concentrated.

Prediction Tab. The Prediction tab presents the model’s 7-day return forecast along with a confidence rating and an uncertainty band. A feature-attribution bar chart decomposes the prediction into its contributing signals—including sentiment strength, discussion volume, subreddit composition, and embedding-derived features—providing transparency into why the model produced the given forecast.

Together, these views form an interactive explanation framework, allowing users not only to see a forecast but also to understand the underlying sentiment patterns, community dynamics, and model-level reasoning that shape the prediction. This multi-layered transparency architecture directly addresses

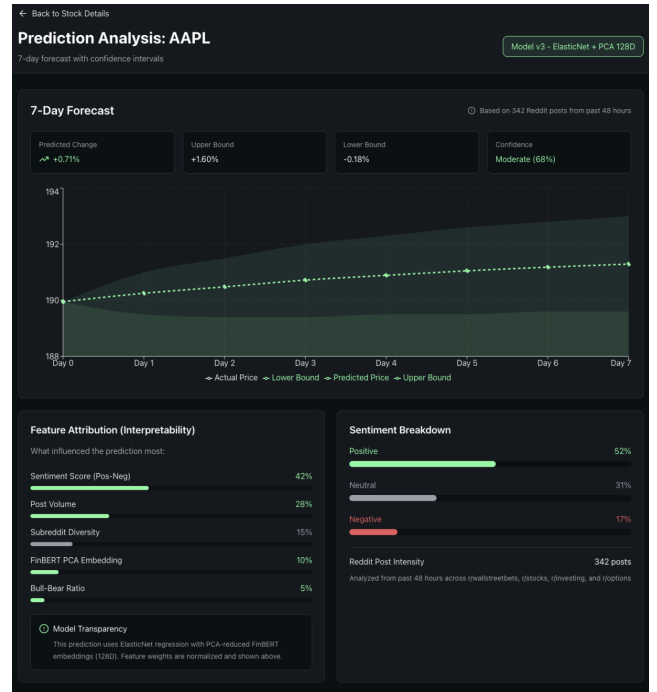


Fig. 6. Prediction tab: model forecast, confidence estimate, and feature-attribution breakdown.

survey findings that users demand visibility into both data sources and model mechanics, with 7 of 10 respondents indicating they would only trust sentiment predictions if source posts and calculation methodologies were exposed.

IV. EVALUATION

Evaluation is performed along two axes: (1) predictive performance of the modeling pipeline, and (2) ecological validity and usability of the prototype.

A. Modeling Evaluation

Model performance was evaluated on the held-out test set using three complementary metrics: Mean Absolute Error

(MAE), the coefficient of determination (R^2), and the Pearson correlation between predicted and actual 7-day price changes. Together, these metrics characterize both absolute prediction accuracy and the model’s ability to capture directional trends in short-horizon returns.

Table II summarizes the performance of all pipeline versions. The v3 model, which incorporates PCA-reduced FinBERT embeddings and engineered interaction features, achieves the strongest results with an MAE of 0.0237, an R^2 of 0.6206, and a correlation of 0.7898. These values represent a substantial improvement over the v2 models, which suffered from high-dimensional noise and limited feature expressiveness.

TABLE II
MODEL PERFORMANCE COMPARISON

Model	MAE↓	R^2 ↑	Corr↑
v3 ElasticNet + PCA(128)	0.0237	0.6206	0.7898
Phase-1 RidgeCV (small dataset)	0.0205	0.3300	0.6900
v2 RidgeCV	0.0413	0.0159	0.1395
v2 LightGBM	0.0412	0.0072	0.1429
Baseline (mean)	0.045	0.0000	0.0000

Predicted vs. Actual Performance: Fig. 7 visualizes predicted versus actual 7-day returns on the test set. The points cluster along a positive diagonal, indicating that the model reliably captures directional movement and relative magnitude even in a noisy financial setting. As expected for short-horizon returns, most values lie near zero, yet the model still demonstrates separation between positive and negative movements. The absence of systematic curvature or asymmetry suggests that the elastic-net estimator is well calibrated and not overfitting localized portions of the feature space.

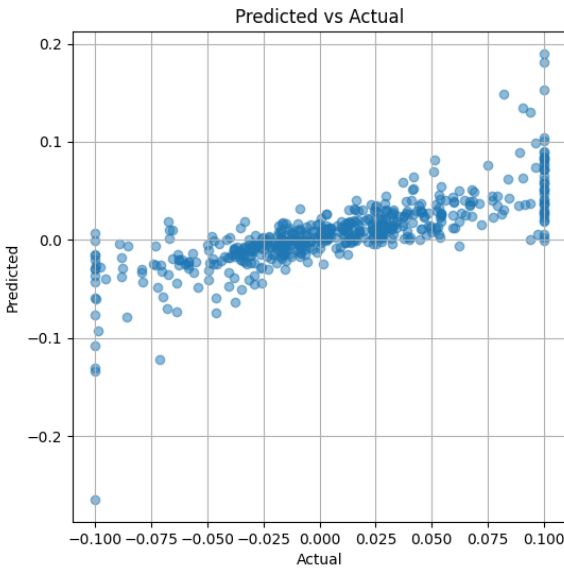


Fig. 7. Predicted vs. actual 7-day returns. The linear trend indicates strong directional alignment.

Residual Analysis: Residual plots (Fig. 8) further validate stability. The residuals are centered tightly around zero with no visible heteroscedastic pattern, indicating that the model does not systematically under- or over-estimate across the prediction range. Larger errors occur primarily at the tails, where financial returns are inherently more volatile. The symmetric, well-dispersed residual cloud confirms that the feature set—particularly PCA-reduced embeddings, temporal encodings, and sentiment-strength interactions—captures the majority of predictable structure available in Reddit posts.

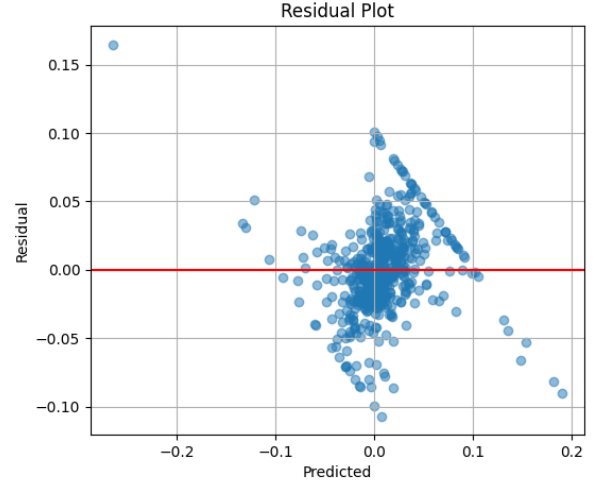


Fig. 8. Residual distribution of the v3 ElasticNet model. Residuals exhibit no structural bias.

Why v3 Outperforms Earlier Versions: Several design choices contribute to the superior performance of the v3 model:

- **PCA for embedding denoising:** Reducing FinBERT embeddings from 768 to 128 dimensions improves signal-to-noise ratio and prevents overfitting.
- **ElasticNet regularization:** The mixture of L_1 and L_2 penalties stabilizes coefficients in a high-dimensional, correlated feature space.
- **Interaction features:** Sentiment–price interactions, sentiment confidence, and bull–bear ratios introduce nonlinear structure that linear models alone cannot capture.
- **Metadata and temporal encodings:** Subreddit, ticker, and cyclical time features add behavioral and contextual information not present in text alone.

Interpretation and Limitations: The results demonstrate that real Reddit discussions contain measurable predictive information about near-term price movement when processed through a structured sentiment and embedding pipeline. However, performance is still limited by the inherent noise of social-media text, uneven posting distribution across tickers, and the modest size of the training corpus. Larger datasets, aggregation at the daily level, and inclusion of historical price features (volatility, momentum) are promising next steps for improving predictive robustness.

Overall, the evaluation confirms that multimodal sentiment modeling—combining FinBERT embeddings, sentiment scores, metadata, and engineered interactions—provides a meaningful directional signal for short-horizon stock forecasting.

B. Prototype Evaluation

The prototype was evaluated through surveys ($n=10$) and a structured interview with a frequent trader who actively uses social media platforms such as StockTwits for investment decisions, providing complementary insights into design priorities and trust mechanisms.

1) *Survey Insights and Design Alignment:* Survey respondents rated social media’s influence on stock prices between 4–5 on a 5-point scale, acknowledging its impact while expressing concerns about manipulation and false positives. Regarding trust in sentiment-powered predictions, responses divided into three categories: those willing to use it as a supplementary tool, those interested but concerned about bias, and those skeptical of social media’s predictive validity.

Key design requirements from survey responses included: (1) transparency through access to source posts rather than only aggregated scores, (2) integration with traditional technical indicators, (3) community-level breakdown showing which subreddits drove sentiment scores, and (4) filtering capabilities by sentiment polarity and discussion volume. The prototype addresses these through tri-level sentiment distributions, word clouds, subreddit comparison bars, recent posts sections with sentiment labels, and explicit confidence ratings with feature-attribution visualizations.

2) *Interview Findings: Usability and Trust:* The interview with an experienced sentiment-driven trader revealed specific usability gaps. While information density was appropriate, critical contextual definitions were missing—users wanted tooltips clarifying percentage values and a section highlighting stocks with drastic sentiment changes. The trader emphasized the causality ambiguity problem: uncertainty about whether sentiment drives prices or prices generate sentiment. The prototype’s temporal sentiment visualization partially addresses this, but additional explanatory guidance is needed.

The “Recent Posts” section validated transparency as essential for trust-building. The trader emphasized verifying aggregated scores against source material and requested clear definitions of score calculation methods. For confidence calibration, the trader wanted both mechanistic explanations of contributing features and historical accuracy rates for confidence tiers. The prototype provides feature attribution but lacks empirical validation metrics showing backtest performance.

3) *Design Strengths and Future Plans:* Key strengths include: three-tab information hierarchy separating exploration from decision-making, explainability-first visualizations prioritizing interpretability, familiar financial dashboard conventions, and multi-granularity analysis from portfolio to post level.

Near-term improvements identified include: tooltip definitions, “most drastic changes” section, historical accuracy

displays for confidence tiers, and actual-versus-predicted price overlays. Future work will conduct comprehension testing with quiz-style checks, trust calibration studies using eye-tracking during simulated trading, A/B testing of interface variants, and longitudinal studies tracking how users integrate sentiment signals into workflows. Recruiting diverse populations including professional traders, quantitative analysts, and retail investors will ensure the interface serves multiple stakeholder groups.

V. CONCLUSION

A. Discussion

This work demonstrates that Reddit sentiment contains measurable predictive signals for short-term stock price movements, though with important qualifications about generalizability and market regime dependence. The success of relatively simple engineered features—sentiment strength, confidence ratios, bull-bear interactions—over raw high-dimensional embeddings suggests that predictive value lies in aggregate conversation quality rather than subtle linguistic patterns. This aligns with behavioral finance theories that investor psychology drives short-term price deviations, and that these psychological states manifest in measurable conversation dynamics.

The superior performance of ElasticNet over gradient boosting challenges the assumption that complex ensemble methods universally outperform linear models. This result likely reflects overfitting tendencies when training data is limited: high-dimensional text embeddings contain rich information but substantial noise that complex models readily memorize. Dimensionality reduction via PCA followed by regularized linear regression appears to extract stable signals while discarding noise, suggesting that explicit feature engineering guided by domain knowledge outweighs architectural sophistication for sentiment prediction with constrained datasets.

User feedback revealed a critical tension between predictive accuracy and trust. Survey respondents expressed skepticism about “black box” forecasts but showed interest in systems exposing underlying posts, sentiment distributions, and explicit confidence levels. This suggests deployment success depends as much on transparency and communication strategy as on model performance—a system with moderate accuracy but high interpretability may prove more valuable than an opaque but accurate alternative. User feedback revealed a critical tension between predictive accuracy and trust. Survey respondents (rating social media influence 4–5 on a 5-point scale) acknowledged sentiment’s market impact while expressing concerns about manipulation and false positives. Interview participants who actively use platforms like StockTwits for trading decisions emphasized that they would verify aggregated scores against source material rather than blindly following metrics. This confirms that deployment success depends as much on transparency and communication strategy as on model performance—a system with moderate

accuracy but high interpretability may prove more valuable than an opaque but accurate alternative.

Several limitations constrain interpretation. The dataset's three-month window and focus on large-cap technology stocks limit understanding of how sentiment-price relationships evolve across market regimes and company characteristics. The relatively stable market conditions during data collection leave open questions about performance during volatility spikes, corrections, or meme episodes. User survey responses (n=10) provide directional insights but insufficient statistical power for definitive conclusions about interface preferences. These constraints suggest our results validate technical feasibility and directional predictive value while leaving substantial questions about practical deployment under varied conditions.

B. Future Work

Several extensions would substantially advance this work's scientific contribution and practical utility. Expanding the dataset to 12–24 months would enable regime-switching models that adapt predictions based on market conditions, distinguishing between normal periods where fundamentals dominate and meme episodes where social coordination drives prices. Incorporating conversation-level features beyond post-level sentiment—thread depth, argument similarity, sentiment coherence across comment chains—would capture coordination dynamics that prior literature suggests drive abnormal returns. Multi-platform integration combining Reddit, Twitter, StockTwits, and financial news would provide complementary signals with different latency and user demographics, though careful fusion methodology would be required to reconcile conflicting signals and platform-specific biases. Real-time deployment through streaming APIs and production web infrastructure would enable live validation and user testing beyond simulated backtesting, revealing performance degradation from market adaptation and changing social media dynamics.

Comparative analysis against traditional technical indicators and fundamental signals would contextualize sentiment's predictive value-add, while ensemble approaches combining social and traditional features could improve accuracy and provide fallback predictions for stocks with insufficient social media coverage. Explainability enhancements through SHAP values or LIME would quantify which posts drive individual predictions, building user trust through mechanistic transparency. Adversarial robustness testing against coordinated manipulation and bot-generated content would reveal vulnerabilities requiring defensive measures before production deployment.

Finally, structured interviews with professional traders and quantitative analysts would provide deeper insights than survey responses alone. Current feedback (n=10 survey, n=1 interview with active StockTwits user) identified specific

interface gaps including tooltip definitions, historical accuracy displays for confidence tiers, and actual-versus-predicted overlays, but larger samples across diverse expertise levels are needed. A/B testing of interface variants would optimize for user comprehension and trust—for example, testing whether numerical confidence scores versus categorical labels (High/Moderate/Low) better calibrate user expectations. Longitudinal studies tracking how users integrate sentiment predictions with existing workflows would reveal practical adoption barriers and inform iterative design improvements. These extensions collectively address current limitations while advancing understanding of social media's role in modern financial markets.

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